D07 · Pick-Pack-Ship & Fulfillment Execution

05 · Fulfillment KPIs: perfect order, DPMO & order cycle time

Supply Chain Reference Architecture v2.2 · mrsupplychain.ai

1 WHAT IT IS

Fulfillment KPIs measure the quality and speed of pick-pack-ship: perfect order rate, DPMO (defects per million opportunities), order cycle time, pick accuracy and cost per order, tracked by process stage.

2 WHY IT MATTERS

Internal efficiency metrics such as lines per hour say little about what the customer received. **Perfect order rate multiplies the failure points — completeness, timeliness, condition, documentation — so it exposes problems that individual metrics hide.** DPMO adds precision for operations where even rare errors are unacceptable, such as pharmaceuticals.

3 HOW IT IS DONE

1. **Define defects and opportunities.** What counts as an error at each stage, and the denominator for each.
2. **Measure perfect order end to end.** Complete, on time, undamaged, correctly documented — all four at once.
3. **Break cycle time into stages.** OMS, pick, pack, ship and transit, to find the slowest link.
4. **Automate the dashboard.** Feed it directly from WMS, TMS and OMS; review monthly with actions.

IN PRACTICE

Perfect order rate is the most useful fulfillment metric for management — not LPPH

LPPH measures internal picking efficiency. Perfect order rate measures whether the customer received what we promised. Only perfect order rate connects fulfillment performance to customer experience.

Calculate DPMO for operations with high quality requirements

For pharmaceutical, precision electronics, or very strict SLA customer operations, DPMO is the correct quality metric. A DPMO of 5,000 in a pharmaceutical operation is unacceptable; in FMCG distribution it may be the starting point.

Order cycle time measures the entire chain's speed — not just the warehouse

A long order cycle time can have its bottleneck in OMS order processing time (before the WMS), carrier transit time (after the WMS), or the picking/packing/shipping process. Stage-by-stage analysis identifies the slowest link.

Automate fulfillment KPIs — don't build them manually

Fulfillment KPIs must be calculated automatically from the WMS, TMS, and OMS. A manually built fulfillment dashboard has a days-long lag and error risk — the opposite of what a continuous improvement process needs.

ILLUSTRATIVE CASE

Illustrative case built from typical industry values — not data from a specific company.

Monthly fulfillment dashboard — B2B distribution DC, October management review — Cycle summary. All metrics vs. target and vs. previous month.

Fulfillment KPI	Current value	Status vs. target
Pick accuracy % (error-free picking)	MEDIUM	99.2% vs. target >99.5% — improving from 98.8% previous month
LPPH (lines per hour per operator)	MEDIUM	78 LPPH vs. target >85 LPPH — re-slotting opportunity detected

Fulfillment KPI	Current value	Status vs. target
Perfect order rate % (complete + on-time + no damage)	HIGH	88.6% vs. target 94% — limited by carrier OTIF (92.4%)
Order cycle time (capture -> delivery, business hours)	OK	5.8 hours vs. target <6 hours — within target

Result: Narrative: Pick accuracy improving but LPPH still below target — top 40 velocity SKUs detected to have migrated from the golden zone in the last 3 months. Re-slotting scheduled for week 3. Perfect order rate low due to carrier OTIF (92.4% vs. 96% target) — action plan with carrier under review.

HOW IT IS MEASURED

Perfect Order Rate % (correct, complete, on-time, no damage, correct documentation)

Perfect Order Rate %(Orders meeting all 5 criteria / Total orders for the period) × 100

Benchmark: >95% perfect order rate in B2B distribution · >99% in pharmaceutical and high-precision electronics Perfect order rate is the most complete end-to-end quality metric. An excellent LPPH with a high damage rate or incorrect documentation results in a low perfect order rate that impacts the customer.

Fulfillment Cost per Order (total fulfillment cost per order)

Fulfillment Cost per Order(Picking + packing + shipping labor cost + materials) / Total orders for the period

Benchmark: $4–12 per order in efficient B2B distribution · $8–25 in e-commerce · Varies by lines/order and automation level Fulfillment cost per order is the financial KPI that connects warehouse operational efficiency with order margin. A $28 fulfillment cost on an order with $22 gross margin destroys value.

WHAT YOU WOULD USE

Fulfillment Analytics

Power BI — Fulfillment KPI Dashboard — Module: Automated Fulfillment ReportingPower BI dashboard with direct WMS, TMS, and OMS connection. Pick accuracy, LPPH, perfect order rate, and order cycle time updated automatically.

SAP EWM Analytics — Module: WMS-Native Fulfillment ReportingNative SAP EWM reports for fulfillment KPIs: per-operator productivity, per-process accuracy, and cycle times by stage.

4 SO WHAT

Anchor fulfillment on what the customer receives. Perfect order end to end, DPMO where precision is critical, cycle time by stage and dashboards fed automatically from the systems.

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